Samarth Agali Prasad

Bengaluru, KA, India | samarthprasad440@gmail.com | 9739253212 | samarthagali.github.io
samarth-agali-prasad | samarthagali

Summary

ML Engineer with experience deploying real world ML systems in fraud detection, RAG-based retrieval systems, and edge inference optimization. Focused on model design and development, scalable deployment, model efficiency, and MLOps workflows.

Experience

Software Development Engineer — *Bengaluru, KA, India*
Craft Silicon Pvt Ltd — *Jan 2025 – Present*

- Built and deployed a RAG-based internal chatbot used by the MIS team, delivering borrower insights across 5900 centers.
- Built and deployed a hybrid supervised and unsupervised fraud detection system in FastAPI, generating risk scores across 80k borrower profiles from real-time data.
- Deployed an internal LLM-based assistant capable of answering queries over a large monolithic legacy codebase, improving developer onboarding and product knowledge across 300 employees.
- Automated documentation generation for a previously undocumented product using AI, producing structured reference documentation from scratch.
- Built a Spark-based log compression pipeline (POC) achieving a 14.5x compression ratio at 3.5 MB/s, reducing a 2.73 GB log folder to 188 MB.

Deep Learning Intern — *Bengaluru, KA, India*
Sandlogic Technologies — *Dec 2022 – May 2024*

- Designed and optimized lightweight deep learning models for edge deployment improving inference latency by 1.5x under low compute constraints.
- Assisted in model optimization workflows and deployment using TVM Relay for edge inference, working with models up to AlexNet scale (62M parameters).

Projects

Aquaplast — *PyTorch, Deep Learning, Docker, Streamlit*

- Built a YOLO-based plastic waste detection and forecasting system using live weather inputs achieving above 85 percent accuracy and deployed a working demo for NGO field testing.

Audiolite — *PyTorch, Deep Learning, Docker, Streamlit*

- Developed a neural audio codec with super resolution achieving up to 8x compression while maintaining perceptual quality and deployed as a containerized app.

Diabetes Predictor — *Machine Learning, CI/CD*

- Built a production oriented MLOps workflow with a five model ensemble and automated retraining achieving 0.995 validation precision.

Movie Recommender Bot — *RAG, FAISS, Streamlit, LLMs*

- Built a RAG-based movie recommendation system indexing 25k embeddings with FAISS and validated against 5000 queries through prompt tuning and guardrails.

Skills

Machine Learning: PyTorch, scikit-learn, OpenCV, Computer Vision, RAG, TVM, Pandas, NumPy
Deployment & Tools: FastAPI, Docker, CI/CD, Streamlit, Git
Systems & Programming: Python, C++, C, SQL, x86 Assembly, Linux, Bash

Publications

Image Compression and Retrieval: A Comprehensive Survey — Dec 2023
Coauthor: Samarth Agali Prasad
DOI: 10.17577/IJERTV12IS123101

Education

Dayananda Sagar College of Engineering *2020 – 2024*
B.E in Artificial Intelligence *GPA: 8.86/10.0*